

Module4_Lab3

September 27, 2026

```
[1]: import pandas as pd
df = pd.read_csv("truck.csv")

r = df['x'].corr(df['y'])
print(f"Pearson's r: {r:.4f}")
```

```
-----
KeyError                                Traceback (most recent call last)
File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:3641, in Index.
    ↪get_loc(self, key)
      3640 try:
-> 3641     return self._engine.get_loc(casted_key)
      3642 except KeyError as err:

File pandas/_libs/index.pyx:168, in pandas._libs.index.IndexEngine.get_loc()
--> 168 'Could not get source, probably due dynamically evaluated source code.'

File pandas/_libs/index.pyx:197, in pandas._libs.index.IndexEngine.get_loc()
--> 197 'Could not get source, probably due dynamically evaluated source code.'

File pandas/_libs/hashtable_class_helper.pxi:7668, in pandas._libs.hashtable.
    ↪PyObjectHashTable.get_item()
-> 7668 'Could not get source, probably due dynamically evaluated source code.'

File pandas/_libs/hashtable_class_helper.pxi:7676, in pandas._libs.hashtable.
    ↪PyObjectHashTable.get_item()
-> 7676 'Could not get source, probably due dynamically evaluated source code.'

KeyError: 'x'
```

The above exception was the direct cause of the following exception:

```
KeyError                                Traceback (most recent call last)
Cell In[1], line 4
      1 import pandas as pd
      2 df = pd.read_csv("truck.csv")
      3
----> 4 r = df['x'].corr(df['y'])
```

```

5 print(f"Pearson's r: {r:.4f}")

File ~\anaconda3\Lib\site-packages\pandas\core\frame.py:4378, in DataFrame.
-> __getitem__(self, key)
4374
4375     if is_single_key:
4376         if self.columns.nlevels > 1:
4377             return self._getitem_multilevel(key)
-> 4378         indexer = self.columns.get_loc(key)
4379         if is_integer(indexer):
4380             indexer = [indexer]
4381     else:

File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:3648, in Index.
-> get_loc(self, key)
3643     if isinstance(casted_key, slice) or (
3644         isinstance(casted_key, abc.Iterable)
3645         and any(isinstance(x, slice) for x in casted_key)
3646     ):
3647         raise InvalidIndexError(key) from err
-> 3648     raise KeyError(key) from err
3649 except TypeError:
3650     # If we have a listlike key, _check_indexing_error will raise
3651     # InvalidIndexError. Otherwise we fall through and re-raise
3652     # the TypeError.
3653     self._check_indexing_error(key)

KeyError: 'x'

```

```

[2]: import pandas as pd
df = pd.read_csv("truck.csv")

r = df['X'].corr(df['Y'])
print(f"Pearson's r: {r:.4f}")

```

Pearson's r: 0.9985

Pearson's r is 0.9985. Not quite a +1 but close! It indicates that there is a strong linear relationship between the list price and the best price.

```

[3]: df.plot.scatter(x, y)

```

```

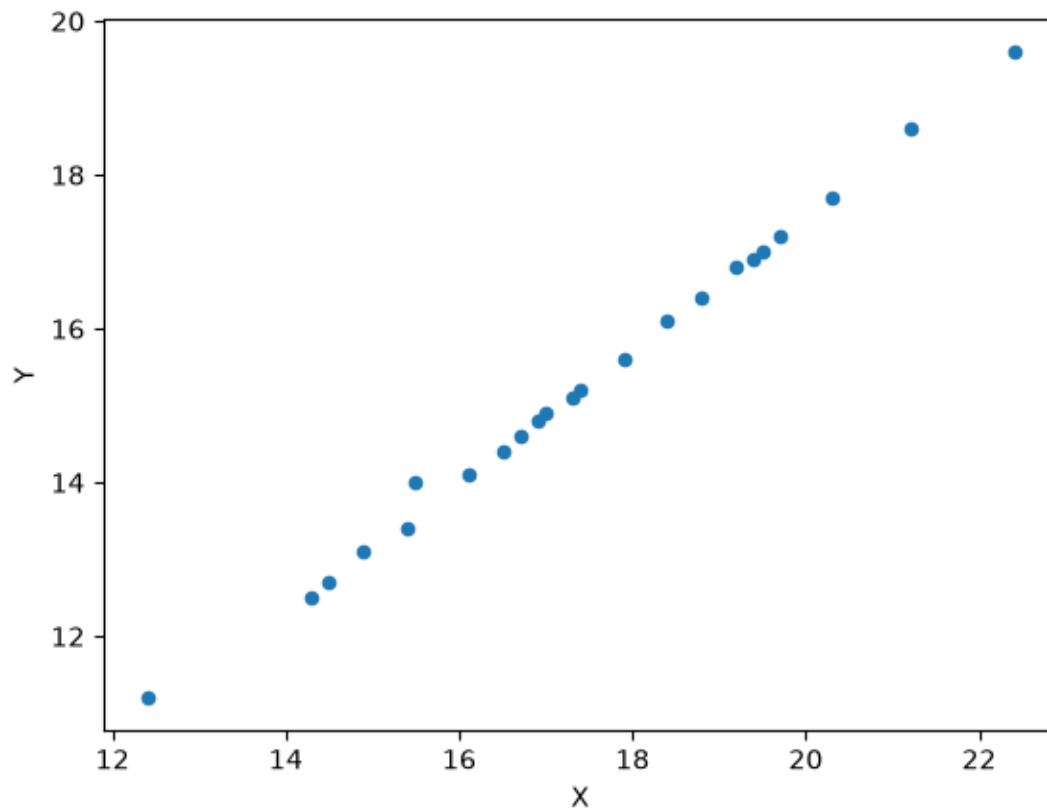
-----
NameError                                Traceback (most recent call last)
Cell In[3], line 1
----> 1 df.plot.scatter(x, y)

```

```
NameError: name 'x' is not defined
```

```
[4]: df.plot.scatter(x='X', y='Y')
```

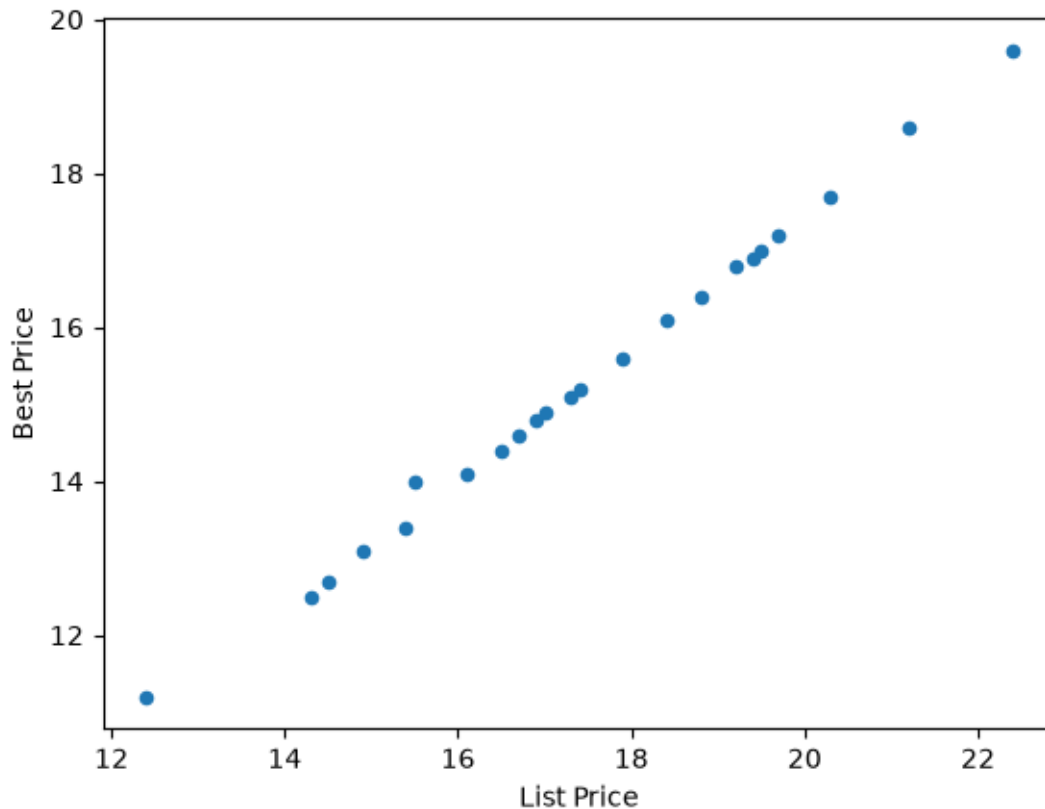
```
[4]: <Axes: xlabel='X', ylabel='Y'>
```



```
[5]: import matplotlib.pyplot as plt
```

```
df.plot.scatter(x='X', y='Y')  
plt.xlabel('List Price')  
plt.ylabel('Best Price')
```

```
[5]: Text(0, 0.5, 'Best Price')
```



```
[6]: slope, intercept, r_value, p_value, std_err = stats.linregress(df['x'], df['y'])
print(f"Regression Equation: Y = {slope:.4f}X + {intercept:.4f}")
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[6], line 1
----> 1 slope, intercept, r_value, p_value, std_err = stats.linregress(df['x'],
    ↪ df['y'])
      2
      3 print(f"Regression Equation: Y = {slope:.4f}X + {intercept:.4f}")

NameError: name 'stats' is not defined
```

```
[7]: from scipy import stats

slope, intercept, r_value, p_value, std_err = stats.linregress(df['x'], df['y'])
print(f"Regression Equation: Y = {slope:.4f}X + {intercept:.4f}")
```

```

-----
KeyError                                Traceback (most recent call last)
File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:3641, in Index.
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    ↪ PyObjectHashTable.get_item()
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```

KeyError: 'x'

The above exception was the direct cause of the following exception:

```

KeyError                                Traceback (most recent call last)
Cell In[7], line 3
      1 from scipy import stats
      2
----> 3 slope, intercept, r_value, p_value, std_err = stats.linregress(df['x'],
    ↪ df['y'])
      4
      5 print(f"Regression Equation: Y = {slope:.4f}X + {intercept:.4f}")

File ~\anaconda3\Lib\site-packages\pandas\core\frame.py:4378, in DataFrame.
    ↪ __getitem__(self, key)
      4374
      4375     if is_single_key:
      4376         if self.columns.nlevels > 1:
      4377             return self._getitem_multilevel(key)
-> 4378         indexer = self.columns.get_loc(key)
      4379         if is_integer(indexer):
      4380             indexer = [indexer]
      4381     else:

```

```

File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:3648, in Index.
-> get_loc(self, key)
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    3644         isinstance(casted_key, abc.Iterable)
    3645         and any(isinstance(x, slice) for x in casted_key)
    3646     ):
    3647         raise InvalidIndexError(key) from err
-> 3648     raise KeyError(key) from err
    3649 except TypeError:
    3650     # If we have a listlike key, _check_indexing_error will raise
    3651     # InvalidIndexError. Otherwise we fall through and re-raise
    3652     # the TypeError.
    3653     self._check_indexing_error(key)

KeyError: 'x'

```

```

[8]: from scipy import stats

slope, intercept, r_value, p_value, std_err = stats.linregress(df['X'], df['Y'])

print(f"Regression Equation: Y = {slope:.4f}X + {intercept:.4f}")

```

Regression Equation: Y = 0.8511X + 0.4346

If $Y = mX + b$

$X = 25.2$

$Y = 0.8511X + 0.4346$

$Y = 0.8511(25.2) + 0.4346$

$Y = 21.882$

```

[9]: m = 0.8511
     x = 25.2
     b = 0.4346

     best_price = 0.8511 * 25.2 + 0.4346
     print(best_price)

```

21.88232

Best price would be \$21,882